

TNA Reconstruction Protocol v1

A Generative Procedure for Recovering and Expanding TNA-Compatible Knowledge Spaces

0. Purpose

This protocol defines a **constructive method** to:

- reconstruct latent theoretical structures (e.g., the “150 papers”)
- generate new valid extensions of TNA
- ensure convergence toward structurally consistent knowledge

It transforms TNA from:

*a descriptive meta-framework
into a **generative epistemic engine***

1. Core Principle

All reconstruction is driven by the canonical law:

boundary → indeterminacy → pressure → expansion

Reconstruction is not retrieval.

It is **forced re-generation through boundary interaction**.

2. Input Specification

Minimal required input:

System S:

type: [physics | ml | cognition | complex]

Ω_0 : initial domain hypothesis

Constraints: [...]

Optional:

Seed knowledge:

known_models: [...]

known_failures: [...]

3. Reconstruction Loop

The system evolves through iterative structural probing:

Step 1 — Domain Initialization

Define:

- admissible space Ω_0
- initial reliability function $R(x)$

Constraint:

- Ω_0 must be **non-trivial** (A1)
 - boundaries must be **detectable** (A2)
-

Step 2 — Boundary Detection

Identify $\partial\Omega$ via:

- instability
- contradiction
- non-generalization
- undefined behavior

Output:

$$\partial\Omega = \{x \mid R(x) \downarrow \text{ or undefined}\}$$

Step 3 — Indeterminacy Mapping

For each boundary point:

- compute ambiguity
- measure non-uniqueness of continuation

Define:

$$I_B(x) = \text{degree of indeterminacy}$$

Step 4 — Pressure Computation

Aggregate boundary effects:

$$F_B = \sum I_B(x)$$

Interpretation:

- high $F_B \rightarrow$ structural tension
 - low $F_B \rightarrow$ locally stable domain
-

Step 5 — Expansion Generation

When:

$$F_B > \tau$$

Trigger:

$$\Omega \rightarrow \Omega^+$$

Generation methods:

- parameter extension
 - rule relaxation
 - domain projection
 - operator introduction
-

Step 6 — Candidate Filtering

For each candidate Ω_i :

Evaluate:

$$R_i(x)$$

Keep only:

- structurally consistent
 - non-degenerate
 - information-preserving extensions
-

Step 7 — Boundary Reformation

For each accepted Ω_i :

- compute new $\partial\Omega_i$
- verify:

$$\partial\Omega_i \neq \emptyset$$

(Prevents false closure)

Step 8 — Iteration

Repeat:

$$\Omega \leftarrow \Omega_i$$

Loop condition:

- indefinite (non-terminal evolution)
-

4. Reconstruction Modes

4.1 Recovery Mode (Backward Reconstruction)

Goal:

- approximate missing theories

Method:

- bias expansions toward:
 - minimal deviation from Ω_o
 - maximum increase in R
-

4.2 Exploratory Mode (Forward Generation)

Goal:

- generate new TNA-consistent theories

Method:

- maximize:
 - boundary diversity
 - expansion novelty
-

4.3 Stress Mode (Boundary Forcing)

Goal:

- accelerate discovery

Method:

- artificially increase F_B via:
 - constraint violation
 - adversarial inputs
 - extreme parameterization
-

5. Output Schema

Each iteration produces:

```
Σ:  
  Ω: domain definition  
  ∂Ω: boundary set  
  R: reliability function  
  H_s: expansion metric  
  F_B: boundary pressure  
  transitions: [...]
```

6. Convergence Criteria

Reconstruction is considered successful when:

- recurrent structures appear across iterations
- cross-domain mappings stabilize
- boundary patterns become invariant

Not:

- when a “final theory” is reached
(TNA forbids terminal closure)

7. Failure Modes

F1 — Boundary Collapse

- $\partial\Omega$ not detected
→ domain overextended
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F2 — Artificial Boundary

- $\partial\Omega$ forced by bad constraints
→ false reconstruction
-

F3 — Expansion Explosion

- uncontrolled Ω growth
→ loss of structure
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F4 — Proxy Degeneracy

- R, F_B not measurable
→ non-operational system
-

8. Minimal Algorithm (IA-Executable)

```
Initialize  $\Omega$ 

while True:

     $\partial\Omega$  = detect_boundary( $\Omega$ )

     $I_B$  = measure_indeterminacy( $\partial\Omega$ )

     $F_B$  = aggregate( $I_B$ )
```

```
if  $F_B > \tau$ :  
    candidates = generate_expansions( $\Omega$ )  
  
     $\Omega = \text{select\_best}(\text{candidates}, R)$   
  
 $\log(\Omega, \partial\Omega, F_B)$ 
```

9. Key Insight

This protocol implies:

The 150 papers are not required to reconstruct TNA.

Because:

- they are **instances of expansions**
 - not the **generator of expansions**
-

10. Canonical Statement

*A theory is not recovered by memory,
but by reproducing the pressure that made it necessary.*

11. Status

This protocol upgrades TNA into:

- a **generative formal system**
- a **search procedure over theory space**
- an **IA-compatible epistemic engine**